NIAGARA 4 SUPERVISOR & JACE | OFFICIAL STEP-BY-STEP USER GUIDE

Ignitechs SMS Notification Driver (ignitechsSms-rt)

Open-Source Internet SMS Platforms, Point Monitoring Rules, Wire-Sheet Triggers & Alarm Routing

Module: ignitechsSms-rt.jar **Vendor: Ignitechs** **Version: 1.0.0** **Palette Symbol: isms** **Target: Niagara 4.0 - 4.14+**

1. Architecture & Component Overview

The Ignitechs SMS Notification Driver (ignitechsSms-rt) adds native internet and network SMS alerting to Niagara 4 Supervisor and JACE stations without requiring any serial port modem hardware. All messages pass through a single background queue with priority ordering, automatic multi-gateway failover, retry scheduling, hourly rate limiting, and duplicate suppression.

```
Station (station:|slot:/)
 +-- Drivers
 |    +-- SmsNetwork (isms:SmsNetwork)
 |         +-- TextBeeCloud      (isms:HttpGateway - Priority 1 Primary Internet SMS)
 |         +-- NtfyInternetAlert (isms:HttpGateway - Priority 2 Backup Mobile Push)
 |         +-- Contacts          (isms:SmsContactFolder - Phone numbers & On-Duty hours)
 |         +-- Groups            (isms:SmsGroupFolder - Named teams e.g. Maintenance)
 |         +-- Rules             (isms:SmsRuleFolder - Point rules & Wire-sheet senders)
 +-- Services
      +-- AlarmService
           +-- SmsAlarmRecipient (isms:SmsAlarmRecipient - Niagara AlarmClass router)
```

2. Step 1 - Install the Module & Trust the Certificate

Step 1.1: Install ignitechsSms-rt.jar into Niagara 4

1. Copy D:\PYTHON\SMSDRIVER\dist\ignitechsSms-rt.jar to C:\Niagara\Niagara-4.14.0.162\modules\ignitechsSms-rt.jar (already installed on this PC, or run .\build.ps1 -Install as Administrator).
2. Restart Niagara Workbench and your Niagara Station so the runtime module is loaded.

Step 1.2: Import the Ignitechs Root CA Certificate

1. In Niagara Workbench, open Tools -> Certificate Management from the top menu bar.
2. Click the User Trust Store tab (and Platform / Station Trust Store if connecting to a remote Supervisor or JACE).
3. Click Import at the bottom of the dialog, select D:\PYTHON\SMSDRIVER\certs\IgnitechsRootCA.pem (or IgnitechsRootCA.cer), and click OK.

Step 1.3: Open the ignitechsSms Palette in Workbench

1. In Niagara Workbench, open the Palette side bar (Window -> Side Bars -> Palette).
2. Click the Open Palette folder icon in the Palette toolbar, type ignitechsSms, select ignitechsSms, and click OK.

3. Step 2 - Add & Configure SmsNetwork Under Drivers

Step 2.1: Add SmsNetwork to Drivers

1. In the Nav tree, expand Station (<StationName>) -> Config -> Drivers.
2. Drag SmsNetwork from the root of the ignitechsSms palette and drop it onto Drivers.
3. Double-click SmsNetwork to open its Property Sheet and review its queue & formatting settings:

Property	Default	Purpose
Enabled	true	Master enable for the SMS network and background dispatch queue.
Site Name	(Station Name)	Building/site label used for the {station} placeholder in messages (e.g. Tower A BMS).
Time Format	dd-MMM HH:mm	Timestamp format used by the {time} placeholder in SMS templates.
Max Retries	3	Number of retry cycles after all enabled gateways fail for a message.
Retry Delay	0000h 01m 00s	Wait time before retrying a failed message.
Max Messages Per Hour	120	Safety rate limit against message storms (0 = unlimited; excess messages wait).
Duplicate Suppression	0000h 02m 00s	Identical text to the same phone number inside this window is sent only once.
Gsm Friendly Text	true	Replaces smart quotes and degree symbols (deg) so messages fit in 160-char GSM-7.

4. Step 3 - Set Up an Open-Source Internet SMS Gateway

Expand the Gateways folder in the ignitechsSms palette. Drag one (or more) of the ready-made internet gateways below and drop it onto SmsNetwork.

Option A (Recommended for Real Carrier SMS): TextBeeCloud (textbee.dev - Open-Source MIT)

1. Create a free account at https://textbee.dev (or run the open-source TextBee Docker container) and install the open-source TextBee Android app on any Android phone with an active SIM card.
2. Scan the QR code in the TextBee web dashboard to link the phone, then copy your Device ID and API Key.
3. In Niagara Workbench, drag TextBeeCloud from Palette -> Gateways onto SmsNetwork.
4. Open the Property Sheet of TextBeeCloud: paste your Device ID into Username, and paste your API Key into Password (Url, Headers, and JSON Body are already pre-configured).
5. Click Save. Right-click TextBeeCloud -> Actions -> Send Test SMS, enter your phone number (+971...), and click OK.

Option B (Open-Source Cloud Relay): SmsGateCloud (capcom6/android-sms-gateway - Apache-2.0)

1. Install the open-source SMS Gateway for Android app (github.com/capcom6/android-sms-gateway), select Cloud Server mode (https://api.sms-gate.app), and toggle the server Online.
2. Note the Username and Password displayed on the app home screen.
3. Drag SmsGateCloud from Palette -> Gateways onto SmsNetwork.
4. Open its Property Sheet, enter the app Username and Password (Basic Auth is pre-enabled), and click Save.
5. Right-click SmsGateCloud -> Actions -> Send Test SMS to verify delivery over the internet.

Option C (Zero SIM Card & Zero Cost): NtfyInternetAlert (ntfy.sh - Open-Source Apache-2.0/GPLv2)

1. Install the free open-source ntfy app on Android/iOS (or open https://ntfy.sh/app in a browser) and subscribe to a unique topic name (e.g. ignitechs-supervisor-alerts-2026).
2. Drag NtfyInternetAlert from Palette -> Gateways onto SmsNetwork.
3. In its Property Sheet, set Username to your topic name (ignitechs-supervisor-alerts-2026) and click Save.
4. Right-click NtfyInternetAlert -> Actions -> Send Test SMS - subscribed phones immediately receive a high-priority mobile push alert over the internet without needing any SIM card or carrier account.

Other Open-Source Internet & Network Gateways in Palette -> Gateways

Palette Component	Platform	Property Sheet Configuration
`TraccarSmsCloud`	Traccar SMS Gateway (Open-Source Apache-2.0)	URL is pre-set to https://www.traccar.org/sms/. Paste your Traccar Cloud API Token into Password.
`JasminHttpGateway`	Jasmin SMS Gateway (Open-Source HTTP API)	Replace <jasmin-host> in Url with your Jasmin server IP/host and set Username and Password.
`PlaySmsGateway`	playSMS Webservice (Open-Source GPL)	Replace <server> in Url with your playSMS server, set Username, and set Password = Webservice Token.
`KannelGateway`	Kannel sendsms (Open-Source HTTP)	Replace <kannel-host> in Url with your Kannel host (:13013) and set Username and Password.
`SmppGateway`	SMPP v3.4 Client (Jasmin / Kannel / Telecom)	Set Address (host), Port (2775), Use TLS, System ID, Password, and Sender ID (e.g. BMS).

TIP - Automatic Multi-Gateway Failover

You can drop multiple gateways under SmsNetwork. Set Priority = 1 on your primary gateway (e.g. TextBeeCloud) and Priority = 2 on a backup gateway (e.g. NtfyInternetAlert or SmppGateway). If Gateway 1 fails, SmsNetwork automatically delivers the message via Gateway 2.

5. Step 4 - Add Contacts & Groups (Address Book)

Step 4.1: Add Contacts under SmsNetwork -> Contacts

1. In the Nav tree, expand SmsNetwork -> Contacts.
2. Drag SmsContact from the ignitechsSms palette into Contacts and name it (e.g. ChiefEngineer or DutyTech).
3. Open its Property Sheet and set Phone Number in international format (e.g. +971501234567).
4. Optional: Configure On Duty Days (days of week) and On Duty Time (working hours) when this contact should receive messages.
5. Right-click the contact -> Actions -> Send Test SMS to verify delivery to that person.

Step 4.2: Create Contact Groups under SmsNetwork -> Groups

1. Expand SmsNetwork -> Groups in the Nav tree.
2. Drag SmsGroup from the palette into Groups and name it (e.g. MaintenanceTeam or NightShift).
3. Open its Property Sheet and expand Members: every contact in Contacts appears as a boolean checkbox.
4. Set the checkboxes to true for all contacts belonging to this group and click Save.

6. Step 5 - Configure Point Monitoring Rules (SmsPointRule)

Step 5.1: Add a Point Rule under SmsNetwork -> Rules

1. Expand SmsNetwork -> Rules in the Nav tree.
2. From Palette -> Rules, drag SmsPointRule (or pre-configured HighValueRule, StateRule, or OfflineRule) into Rules.
3. Open the rule Property Sheet, click the folder chooser on Source ORD, and select any Niagara point in the station.
4. Select Condition, threshold (Limit or Compare Value), Active Delay, Repeat Interval, and Deadband.
5. Expand Recipients and set checkboxes to true for the Groups and/or Contacts that should receive this rule's alerts.
6. Click Save. Right-click the rule -> Actions -> Send Test SMS to test it immediately with the live point value.

Rule Property	Description & Options
`Source ORD`	ORD of any Niagara BControlPoint or status/value slot (e.g. station:|slot:/Drivers/BacnetNetwork/AHU1/points/SAT).
`Condition`	Greater than (>), Less than (<), Greater or equal (>=), Less or equal (<=), Outside range, Inside range, Value equals (=), Value not equals (!=), Is true / active, Is false / inactive, Any value change, Status is Alarm / Fault / Down / Overridden / Not OK.
`Compare Value`	For Value equals / not equals: supports pipe or comma lists such as Fault|Trip|Alarm, On, true, Unoccupied.
`Active Delay`	Condition must remain true for this duration before sending an alert (filters short sensor spikes).
`Repeat Interval`	Re-send reminder SMS at this interval while the condition stays active (up to Max Repeats; 0 = send once).
`Alert / Normal Text`	Supports {station}, {time}, {point}, {path}, {value}, {status}, {state}, {rule}, {condition}, {limit}, {low}, {high}, {compare}, {description}.
`Recipients`	Live auto-synced checklist of all Groups (Group: <name>) and Contacts in SmsNetwork.

7. Step 6 - Use Wire-Sheet SmsSender & AlarmService SmsAlarmRecipient

Step 6.1: Trigger SMS from Wire-Sheet Logic (SmsSender)

1. Drag SmsSender from Palette -> Rules onto any Wire Sheet in your station.
2. Link any boolean output slot to the Trigger input of SmsSender (sends Message on false -> true rising edge), or link an action to Send / Send Text.
3. Open SmsSender Property Sheet, customize Message ({station}, {time}, {sender}), and tick Recipients.

Step 6.2: Route Niagara Alarms to SMS (SmsAlarmRecipient)

1. Open Station -> Config -> Services -> AlarmService (Wire Sheet view).
2. Drag SmsAlarmRecipient from the ignitechsSms palette onto the AlarmService Wire Sheet.
3. Link alarmEscalation1 from your AlarmClass (e.g. defaultAlarmClass) to routeAlarm on SmsAlarmRecipient.
4. Open SmsAlarmRecipient Property Sheet, expand Recipients, and set checkboxes to true for the target Groups/Contacts.
5. Customize Alarm Text ({station}, {state}, {source}, {msg}, {alarmClass}, {priority}, {value}, {time}, {ackHint}) and Ack Text.

8. Step 7 - Verification, Diagnostics & Troubleshooting

Open the Property Sheet of SmsNetwork at any time to inspect Queue Size, Total Sent, Total Failed, Last Sent Time, Last Error, and the live 50-line Activity log.

Diagnostic Message	Resolution
`no reachable recipients`	Tick at least one Contact or Group under Recipients, verify Phone Number is valid (+countrycode...), and check On Duty Days / Time.
`HTTP 401 / HTTP 403`	Check Username (Device ID / Account) and Password (API Key / Token) on the HttpGateway Property Sheet.
`suppressed duplicate`	Identical message text was sent to the same number within Duplicate Suppression (default 2 min). Lower Duplicate Suppression on SmsNetwork when testing.
`Sync Recipients action`	If contacts or groups were edited while the station was offline, right-click SmsNetwork -> Actions -> Sync Recipients.